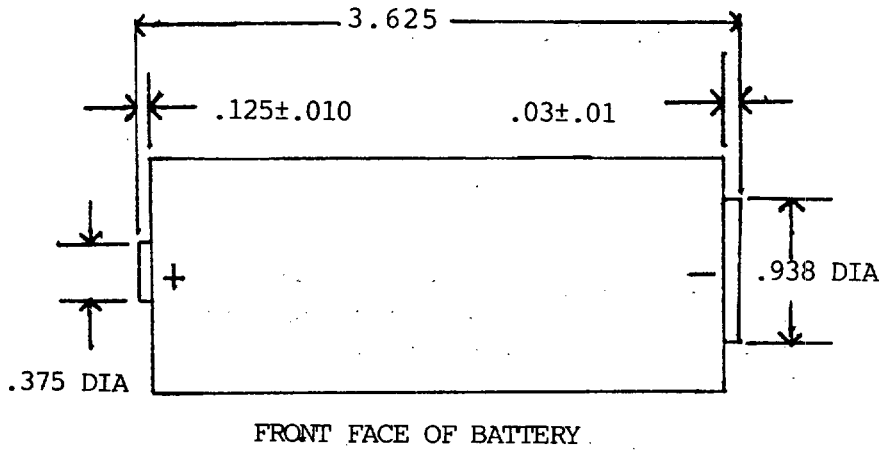
MILITARY SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM THIONYL CHLORIDE BA-6516/U

This specification is approved for use by U.S. Army Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the batteries described herein shall consist of this specification and the latest issue of MIL-B-49461(ER).



INCHES	MM
.010	.25
.015	.38
.030	.76
.031	.78
.125	3.18
.312	7.92
.375	9.53
.938	23.83
1.391	35.33
3.625	92.08

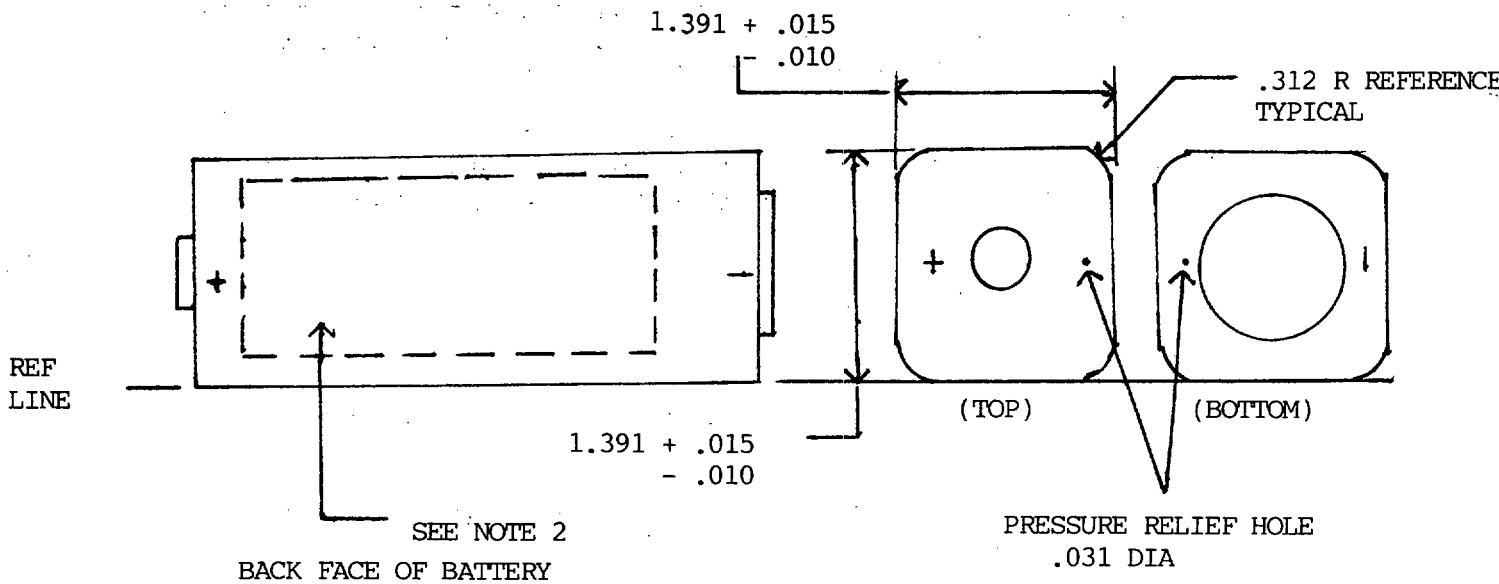


FIGURE 1.0 BA-6516/U BATTERY DIMENSIONS AND LABELS

FSC 6135

AMSC N/A

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Notes

(1) Dimensions are in inches and unless otherwise specified decimal tolerances are ± 0.031 in.

(2) Identification label shall include complete label information required in specification. It shall be located as shown on drawing and surface shall be designated as top of battery. If necessary, adjacent side may contain continuation of label content. The identification label shall contain the following information:

Battery, Non-Rechargeable Lithium Thionyl Chloride, BA-6516/U

(Contract No.)
(Date Code) (Serial No.)
(Manufacturer)
(Location)

(3) Terminal markings. The positive (+) terminal marking may be painted red. The negative (-) terminal may be painted black. The two termianls shall be low resistance flat contacts of nickel per MIL-N-46025.

(4) A. 1.5 ampere slow blow non-replaceable fuze shall be used.

(5) The thermal switch shall be physically located as near as possible to the central most cell. It may be wired at any point in the electrical circuit of the battery.

(6) Design and Construction: The battery shall consist of eight (8) series connected lithium thionyl chloride flat type cells sized for packaging in the battery envelope shown in Figure 1-0.

Electrical Requirements for Battery

Battery Voltage:

No Load Voltage: The no load voltage shall not exceed 30 volts.

Operating Voltage: The nominal operating voltage shall be 24 volts.

End Voltage: The cutoff voltage shall be 20 volts

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Capacity Tests:

(1) Storage: Details on storage conditions for all capacity tests are specified in basic specification.

(2) Discharge: Batteries shall be pulse discharged at 1.5A for 1 second, then off for 9 seconds. This 1 second on 9 seconds off cycle will be repeated continuously until zero volts is reached. However, time to 20V shall be used for calculation of capacity and/or service life. For the I capacity test the batteries shall be pulse discharged at reverse (negative) voltage for 0.7 amp-hours immediately after zero volts is obtained. There shall be no venting, fire or explosion when the batteries are forced discharged.

Capacity Tests	Battery Capacity in Amp-hours to 20.00 volts
I	1.4
L	0.5
H	1.4
HT	1.0
LT	0.3
T	1.4
D	1.4
IT	1.0

Initial Voltage Delay: When the battery is subjected to any of the capacity tests, the time required at the beginning of discharge for the battery to reach a voltage of 20.00 volts after the specified load is applied shall not be more than 1.0 minute.

Voltage Delay: A fixed resistor of 13 ohms shall be applied to the battery prior to the capacity discharge tests to measure voltage delay.

Custodian:
Army - ER

Preparing Activity:
Army - ER
(Project No. 6135-A184-01)

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